Design of Folded Cascode Operational Amplifier in 0.18µm CMOS Technology

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1. Introduction

A folded cascode op-amp has been designed using 0.18 μm CMOS technology. Two different configurations, using a NMOS based and a PMOS based have been designed and their performances compared.

The folded cascode differential op-amps are extensions of telescopic op-amp with slightly lesser gain and avoiding the mirror poles in case of the latter.

Here's a list of specifications being required from the design of 2-stage op-amps:

Gain	> 55 dB
Common-mode input range	0.7 V to 1.1 V
Common-mode output range (Closed loop and Open-loop)	0.5 V to 1.3 V
Common-Mode Rejection Ratio	> 70 dB
Slew Rate	> 10 V/μs
Bandwidth	> 20 MHz
Phase Margin	> 60 º
Gain Margin	> 10 dB

2. Simplified Design of NMOS based 2-stage op-amp

Fig. 1 shows the schematic of a folded cascode op-amp being used for this design.

The design of 2-stage opamp is shown here:

- *Listing all the required parameter values.*

```
> vincmmin:=0.7; vincmmax:=1.1; GB:=22e6; SR:=10e6; vdd:=1.8; vthn:=0.5;
  vthp:=0.6; cl:=5e-12; voutmax:=1.55; voutmin:=0.25;
> eox:=3.45e-13; cox:=7.673e-3; kn:=207.3e-6; kp:=8.466e-5;
```

- *Finding Cc (compensation capacitor). Its value is selected slightly higher than the 2.2 dB margin giving Cc to be around 4/10 times the value of the output load impedance.*

```
> cc:=cl*0.4;
```

$$cc := 0.20\ 10^{-11}$$

Figure 1 Simplified Design of folded-cascode op-amp with 5 pF load capacitance

- *Computing the current required to be sunk by the current source transistor M_5 to ensure the slew rate is high. (Thus, the minimum here is 20 μA)*

> **I5:=SR*cc;**

$$i5 := 0.0000200$$

- *Designing Device size of M_3 and M_4:*

Considering Vsg3=1V

> **Vsg3:=1;**

$$Vsg3 := 1$$

$$s3 := 1.476494212$$

> **l3:=0.3e-6; w3:=s3*l3;**

$$l3 := 0.3\ 10^{-6}$$

$$w3 := 0.4429482636\ 10^{-6}$$

- *Check: Ensuring the mirror pole at the junction of drain of M_1 and drain of M_4 to be a high frequency pole. It can be seen that p3, the pole frequency is ~100 Ghz.*

> **gm3:=kp*s3*Vsg3;**
> **cgs3:=0.667*w3*l3*cox;**
> **p3 := -gm3/(2*cgs3);**

$$gm3 := 0.0001250000000$$

$$cgs3 := 0.6800882795\ 10^{-15}$$

$$p3 := -0.9189983400\ 10^{11}$$

- *Finding the device sizes of M_1 assuming GB = 22e6,*

> **gm1:= (GB)*cc*2*3.142;**

$$gm1 := 0.0002764960$$

```
> s1:=gm1^2/(2*kn*i5/2);
> l1:=0.2e-6; w1:=s1*l1;
> vgs1:=sqrt(i5/(s1*kn))+vthn;
```

$$s1 := 18.43946889$$

$$l1 := 0.2\ 10^{-6}$$

$$w1 := 0.3687893778\ 10^{-5}$$

$$vgs1 := 0.5723337770$$

- *Finding dimensions of M5*

```
> vds5:=vincmmin-vgs1;
> s5:=2*i5/(kn*(vds5)^2);
> l5:=0.3e-6; w5:=s5*l5;
```

$$vds5 := 0.1276662230$$

$$s5 := 11.83882686$$

$$l5 := 0.3\ 10^{-6}$$

$$w5 := 0.3551648058\ 10^{-5}$$

- *Designing second-stage M_6 and M_7:*

```
> gm6:=(2.2*gm1/cc)*cl;
> vsd6:=vdd-voutmax; vds7:=voutmin;
> s6:=gm6/(kp*vsd6);
> l6:=0.2e-6; w6:=l6*s6;  i6:=gm6^2/(2*kp*s6);
```

$$gm6 := 0.001520728000$$

$$vsd6 := 0.25$$

$$vds7 := 0.25$$

$$s6 := 71.85107488$$

$$l6 := 0.2\ 10^{-6}$$

$$w6 := 0.00001437021498$$

$$i6 := 0.0001900910000$$

- *Designing the device size of M_7:*

```
> s7:=s5*i6/i5;
> l7:=0.3e-6; w7:=s7*l7;
```

$$s7 := 112.5227218$$

$$l7 := 0.3\ 10^{-6}$$

$$w7 := 0.00003375681654$$

- *Calculating the gain and power dissipation:*

```
> Pdiss:=vdd*(i5+i6);
> Av:=20*log10(2*gm1*gm6/(i5*2*(0.04+0.05)*i6 ) );
```

$$Pdiss := 0.0003781638000$$

$$Av = 61.79012670$$

- *Estimating the dominant pole frequency, first non-dominant pole and zero:*

```
> P1:=-(0.04+0.05)*i5*i6/(gm6*cc);
> P2:=-(gm6/cl)/(2*3.142);
> Z1:=(gm6/cc)/(2*3.142);
```

$$p1 := -112500.0000$$

$$p2 := -0.4840000001\ 10^{8}$$

$$z1 := 0.1210000000\ 10^{9}$$

Table 1 shows a comparison of the final values obtained theoretically against the experimental values. It can be observed that except the sizes of the active load (M_3 and M_4), the remaining sizes do hold good in the actual experimental values when compared to the final values.

Table I Initial W/L ratios

Parameters	Theoretical Values (in µm)	Simulated Values (in µm)
W_1, L_1 (W_2, L_2)	**3.68, 0.2**	**23,0.3**
W_3, L_3 (W_4, L_4)	**0.44, 0.3**	**80, 0.6**
W_5, L_5	**3.55, 0.3**	**5, 0.6**
W_6, L_6	**14.3, 0.2**	**350, 0.6**
W_7, L_7	**3.3,0.3**	**8, 0.6**

3. Op-amp measurements:

a.) Measuring Input Common-Mode Range (ICMR):

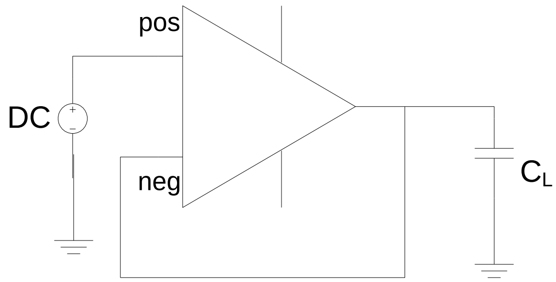


Figure 2 Unity-Buffer Op-amp

To measure the ICMR, the op-amp is connected as shown in Fig. 2. A DC simulation is run with the input DC Voltage swept from 0 to 1.8 V. At the output, Vout/Vin is plotted against Vin. From Fig. 3, it can be seen that the op-amp has a very large ICMR exceeding from ***0.496V to 1.22V*** in the specified range.

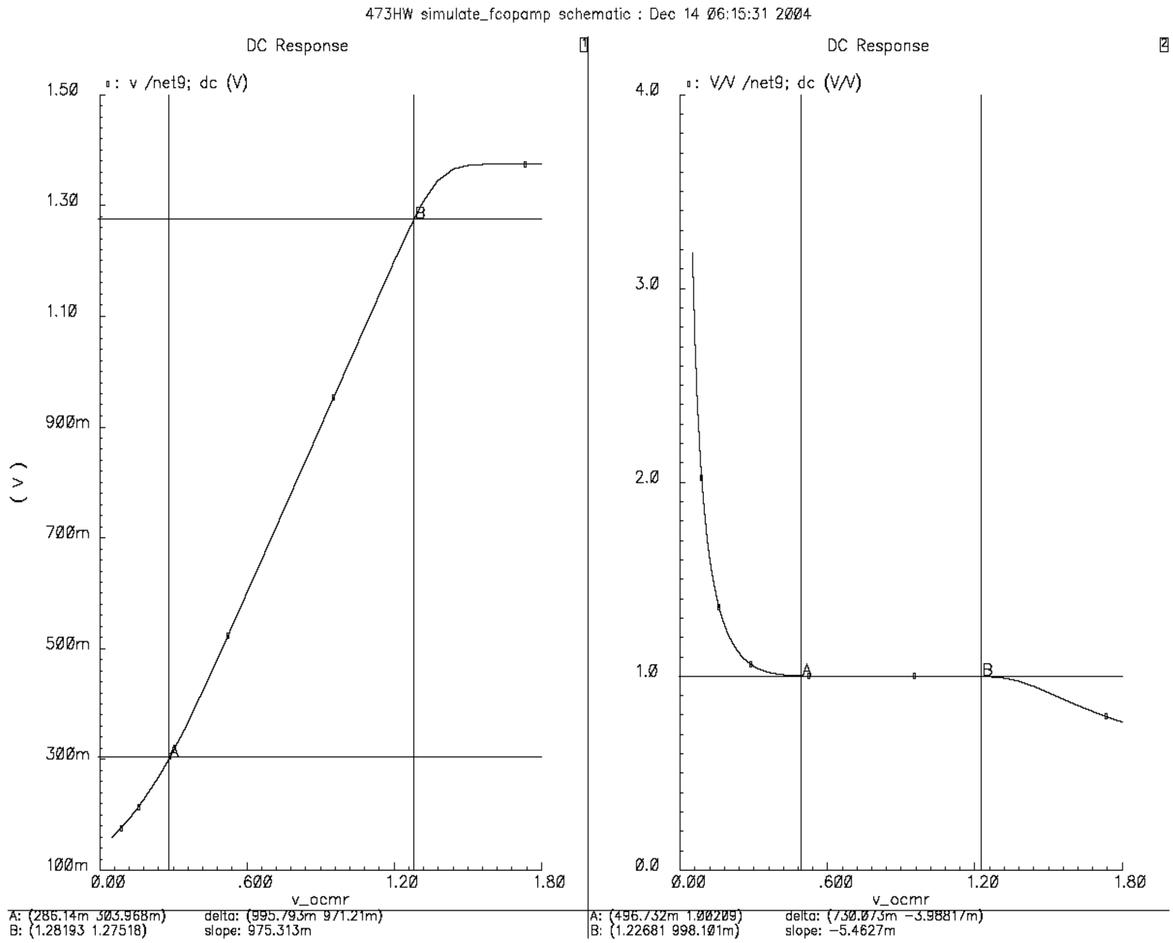


Figure 3 ICMR (Vout/vin vs Vin)

Fig. 4 shows the linear graph between Vout and Vin from which it is hard to decipher the linear region. A second graph shows the current in transistors M_1 and M_5 (current source) indicating that M_5 turns ON at the same time as M_1.

b.) Measuring Output Common-Mode Range (OCMR):

- **Open-loop OCMR:**

To measure the output range, the op-amp is connected as shown in Fig. 5 and the DC value is swept from 0 to 1.8 V. Note that a fixed bias voltage is provided to the negative terminal.

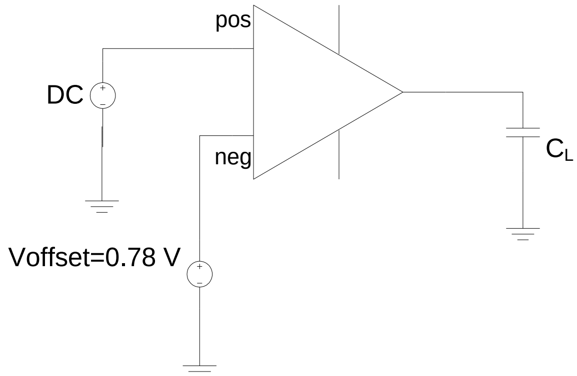


Figure 4 Measuring OCMR by sweeping input "DC" voltage

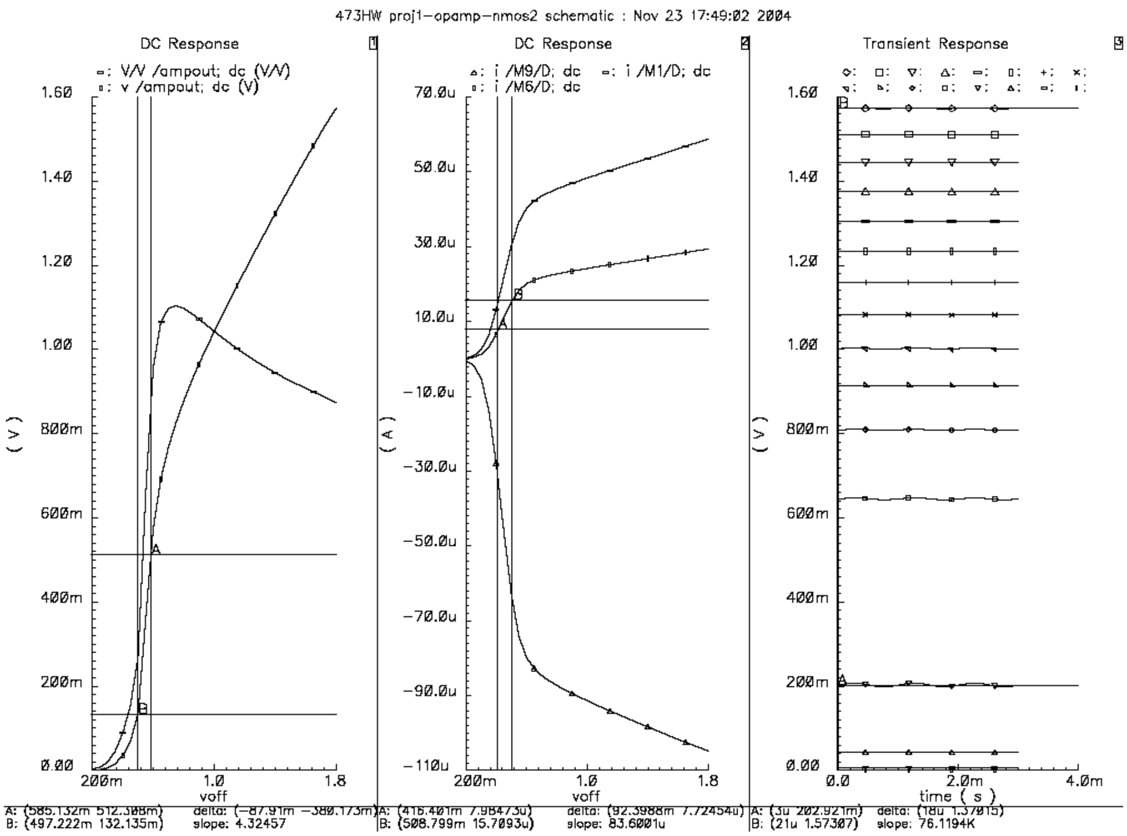


Figure 5 OCMR outputs a.) Vout vs Vin b.) Currents c.) Transient Response of Vout at different Vdc

- **Closed Loop OCMR:**

For closed-loop OCMR, the circuit is configured as shown in the figure 7.

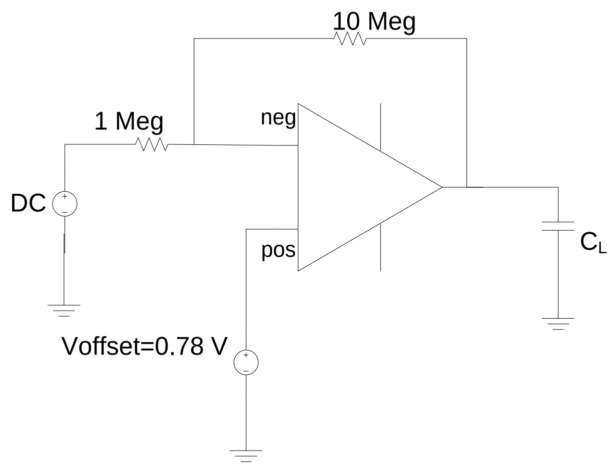


Figure 6 Closed Loop OCMR

Fig. 8 shows the OCMR outputs for open-loop (fig. 8a) and closed-loop configurations (fig. 8b). From Fig. 7b, it can be observed that the output range is from **<u>1.573 to 0.160 V.</u>** To measure open-loop differential gain, bandwidth and phase margin, the

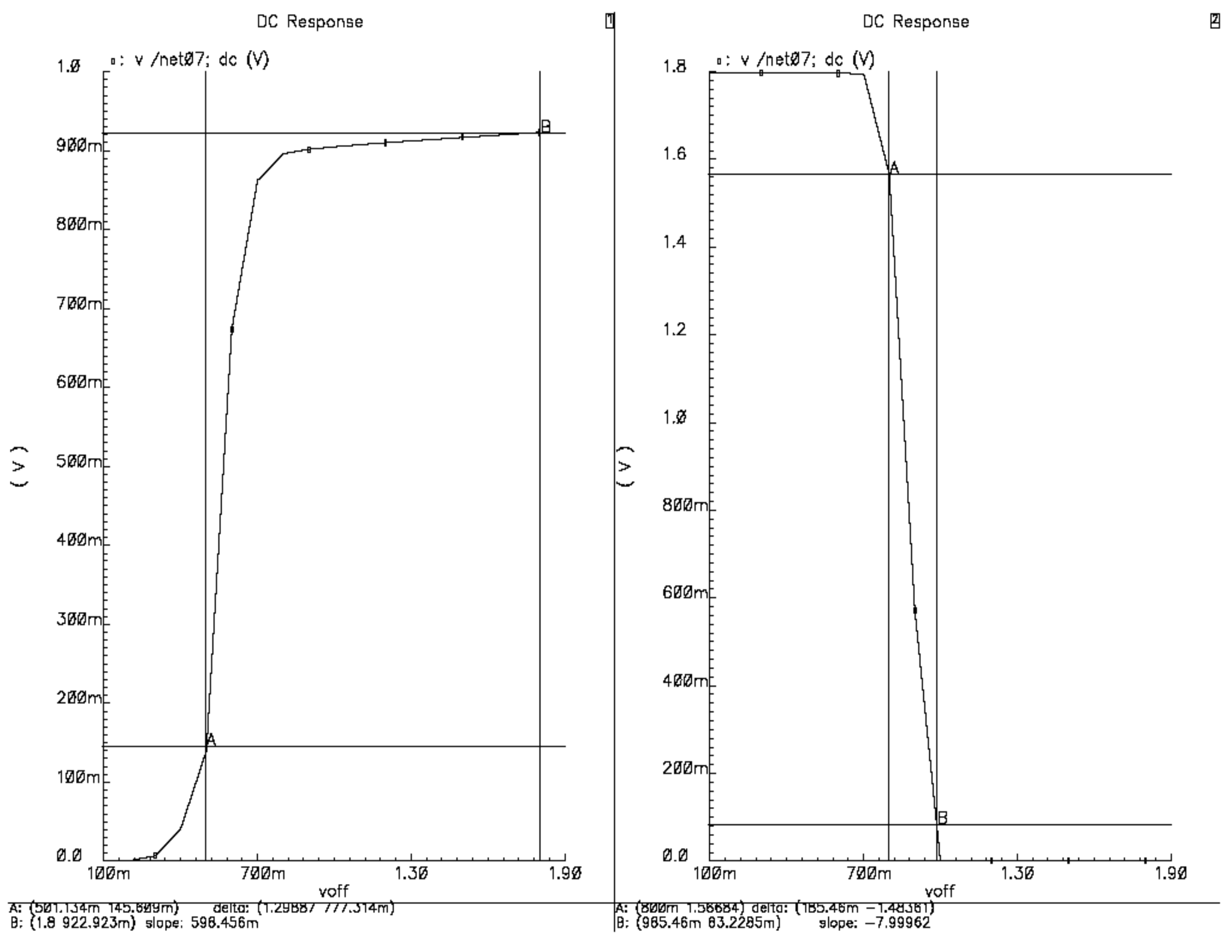


Figure 7 a.) Open Loop OCMR b.) Closed loop OCMR

c.) Measuring Open-loop Differential Gain, Bandwidth and Phase and Gain Margin:

The open-loop differential gain, bandwidth and phase margin are measured from the "AC Gain and phase" plot with the op-amp connected as shown in the Fig. 9. ***The measured gain values are 65.1 dB with Unity Gain Bandwidth of 20.893 MHz, Gain margin of 15.52 dB and Phase margin of 62.67 deg as shown in Fig. 11.***

- ***Use of Miller Compensation to improve Phase margin:***

Miller compensation technique was used to improve the phase margin and closed-loop stability of the op-amp. A nulling resistor was used to further improve the phase margin and the bandwidth by making the zero to cancel the first-nondominant pole in the s-plane. The value of this nulling resistor is designed to be nearly $5/gm_6$ (assuming C_c = 4 pF) which is nearly equal to 3 Kohm.

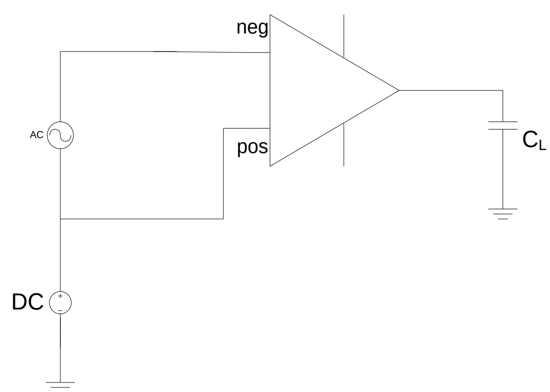


Figure 8 Measuring Open-loop Differential Gain

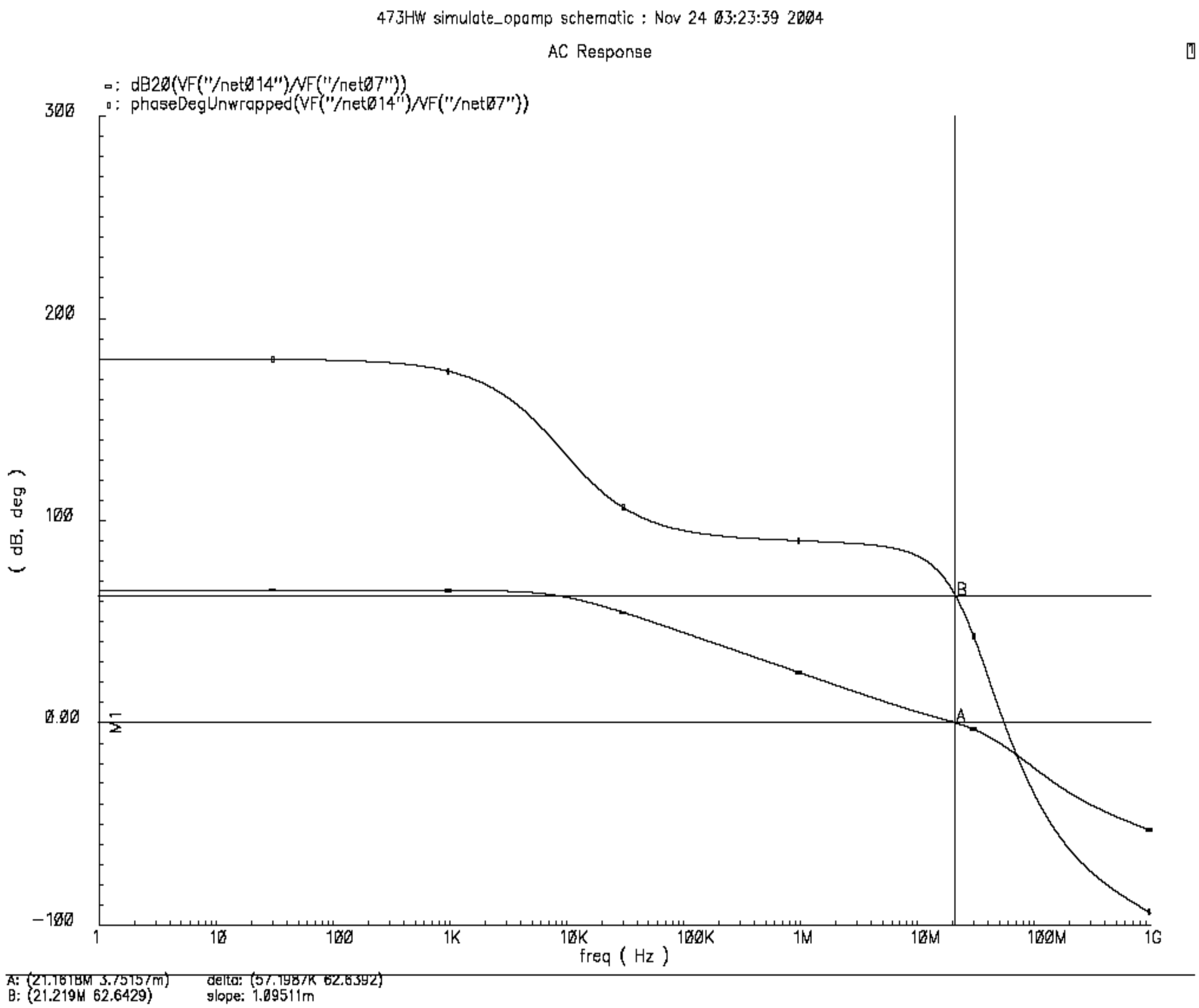


Figure 9 Open loop Differential Gain, BW and Phase Margin

d.) Measuring Power Dissipation:

For measuring the power dissipation, the op-amp is connected as a unity-gain buffer as shown in Fig. 2. It is found that the power dissipation is 1.8 V x 290.8 μA = 0.523 mW.

e.) Measuring Common-Mode Gain/CMRR:

For measuring common-mode gain, the circuit is connected as shown in Fig. 11. Both the inputs are connected together, and the input AC voltage and DC voltage are connected in series. This would yield the common-mode gain. To find CMRR, the common-mode gain is subtracted from the differential gain.

The CMRR is found to be **65 dB** if the configuration in Fig. 1 is used. Hence, the circuit was modified to use a wide-range cascode current source as shown in Fig. 12. The transistor sizes are listed in Table 2. Using this configuration, the CMRR is found to be nearly **95 dB** as shown in Fig. 13. *Please refer the design considerations for the cascade from the PMOS design.*

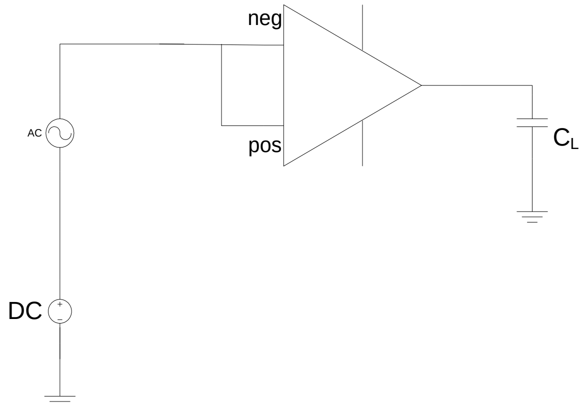


Figure 10 Measuring Common Mode Gain

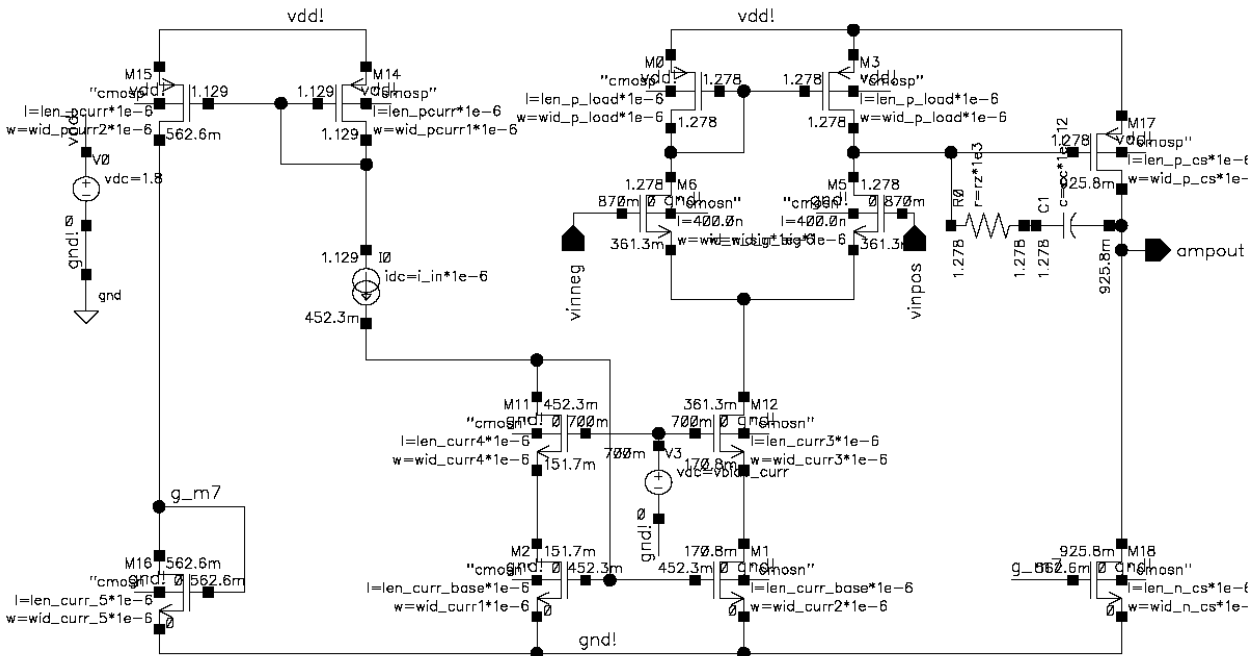


Figure 11 Wide-range Cascode 2-stage Op-amp

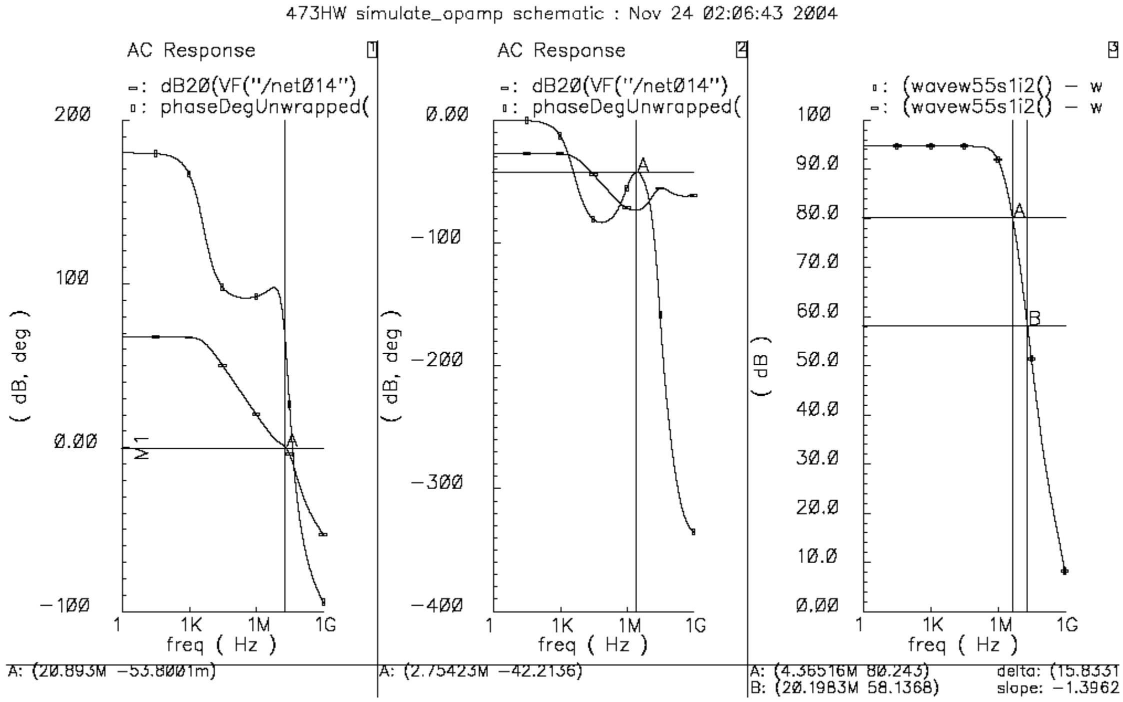


Figure 12 a.) Differential Mode Gain (65.56 dB, BW=20.89 MHz, Phase Margin= 62 deg) b.) Common Mode gain (-26.6 dB) c.) Low Frequency CMRR (>80 dB till 20 MHz)

f.) Measuring Slew Rate:

For measuring the slew rate, the circuit was connected as shown in Fig. 14. A transient simulation is run as shown and the frequency of the sinusoidal source is increased till the output waveform is unable to scale to the input voltage level of 1.1 V (shown in Fig. 15). However, with the new configuration, the negative slew rate is found to be worse equal to **7.85 V/µs** and the positive slew rate is found to be **10.81 V/µs.**

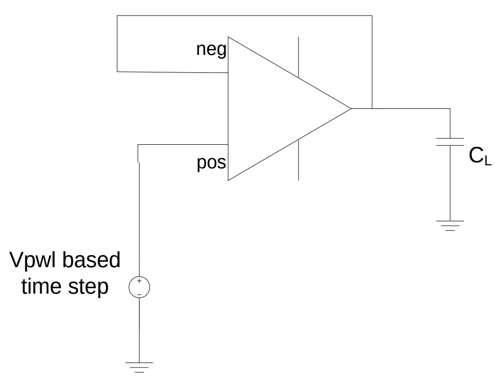


Figure 13 Measuring Slew Rate

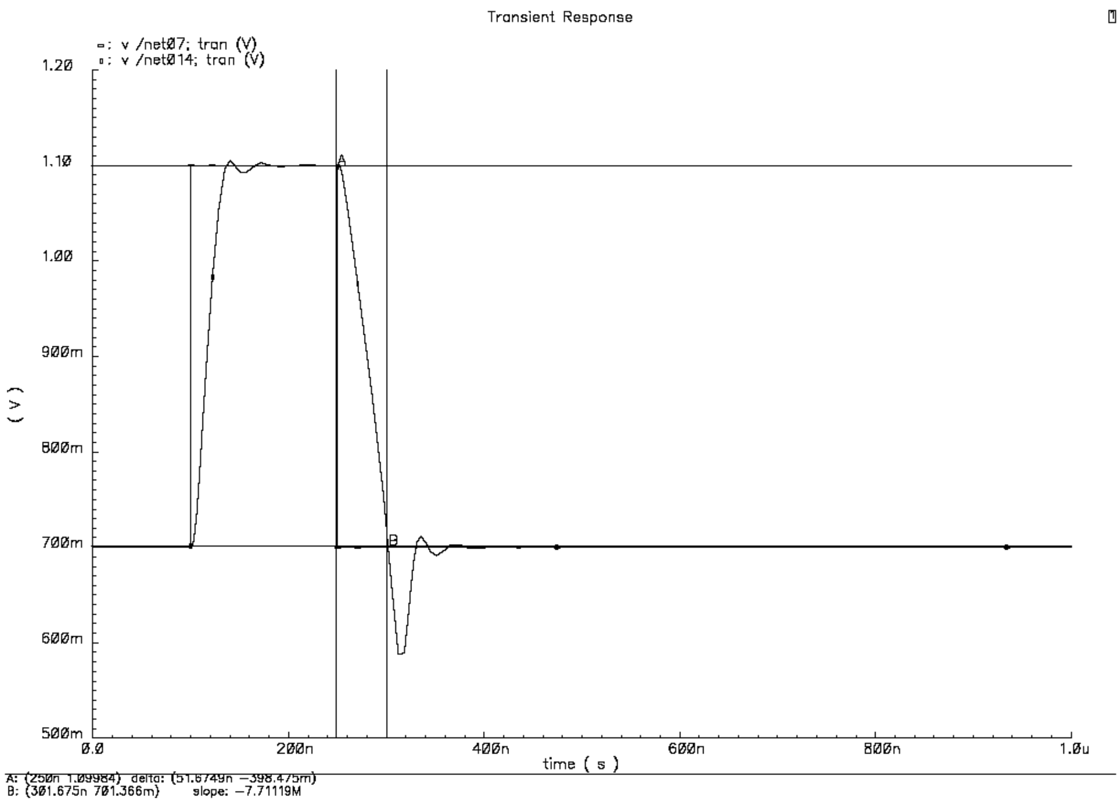


Figure 14 Slew rate measurement - Both positve and negative slew rates are included.

Table III Sizes for op-amp in wide-range cascode configuration

Circuit Elements		**Sizes (in µm)**
Active Load input stage PMOS	M_3 and M_4 PMOS	60, 1.2
Input Signal Path	M_1 and M_2 NMOS	23, 0.3
Second-stage Common source	M_6 PMOS	207, 1.2
Second-stage current source load	M_7 (NMOS)	8, 0.6
First-stage current source NMOS	Cascode Current mirror	(30, 0.6), (20,0.6) (100,0.6),(45,0.6)
Current mirror bias to M_7	NMOS	(10,1.2)
Input to the current mirror bias to M_7	PMOS Current mirror	(25,0.6), (30,0.6)
Compensation cap	Cc	4 pF
Nulling Resistor	Rz	3 K
Vbias_cascade		0.7
Vbias_signal		0.87
Input current		100 µA

PMOS First Pass Hand Calculations

- *Designing Device size of M_3 and M_4:*

Considering Vsg3=1V

```
> Vsg3:=.8v;
> s3:=i5/(kn*(Vsg3-vthp)^2);
```

$$Vgs = .8v$$

$$s3 = 2.41546$$

```
> l3:=0.3e-6; w3:=s3*l3;
```

$$l3 := 0.3\ 10^{-6}$$

$$w3 = .724638 \cdot 10^{-6}$$

- *Check: Ensuring the mirror pole at the junction of drain of M_1 and drain of M_4 to be a high frequency pole. It can be seen that p3, the pole frequency is ~100 Ghz.*

```
> gm3:=kn*s3*Vsg3;
> cgs3:=0.667*w3*l3*cox;
> p3 := -gm3/(2*cgs3);
```

$$gm3 = .000401$$

$$cgs3 = 1.11219\ e\text{-}15$$

$$p3 = -3.6055\ e11$$

- *Finding the device sizes of M_1 assuming GB = 22e6,*

```
> gm1:= (GB)*cc*2*3.142;
```

$$gm1 := 0.0002764960$$

```
> s1:=gm1^2/(2*kp*i5/2);
> l1:=0.2e-6; w1:=s1*l1;
> vgs1:=sqrt(i5/(s1*kp))+vthp;
```

$$s1 = 45.1512$$

$l1 := 0.2\ 10^{-6}$

w1= 9 e-6

vgs1 = .82874

- *Finding dimensions of M5*

```
> vds5:=vdd-vincmmax-vgs1;
> s5:=2*i5/(kp*(vds5)^2);
> l5:=0.3e-6; w5:=s5*l5;
```

vds5 = -.12874

s5 = 40

$l5 := 0.3\ 10^{-6}$

w5 = 12 e-6

- *Designing second-stage M_6 and M_7:*

```
> gm6:=(2.2*gm1/cc)*cl;
> Vsd7:=vdd-voutmax; vds6:=voutmin;
> s6:=gm6/(kn*vsd6);
> l6:=0.2e-6; w6:=l6*s6;  i6:=gm6^2/(2*kn*s6);
```

$gm6 := 0.001520728000$

$vsd6 := 0.25$

$vds7 := 0.25$

s6 = 29.3855

$l6 := 0.2\ 10^{-6}$

w6 = .000006

$i6 := 0.0001900910000$

- *Designing the device size of M_7:*

```
> s7:=s5*i6/i5;
```

\> **l7:=0.3e-6; w7:=s7*l7;**

$$s7 = 380.18$$

$$l7 := 0.3\ 10^{-6}$$

$$w7 = .00000114054$$

- *Calculating the gain and power dissipation:*

\> **Pdiss:=vdd*(i5+i6);**
\> **Av:=20*log10(2*gm1*gm6/(i5*2*(0.04+0.05)*i6));**

$$Pdiss := 0.0003781638000$$

$$Av = 61.79012670$$

- *Estimating the dominant pole frequency, first non-dominant pole and zero:*

\> **P1:=-(0.04+0.05)*i5*i6/(gm6*cc);**
\> **P2:=-(gm6/cl)/(2*3.142);**
\> **Z1:=(gm6/cc)/(2*3.142);**

$$p1 := -112500.0000$$

$$p2 := -0.4840000001\ 10^{8}$$

$$z1 := 0.1210000000\ 10^{9}$$

Table 1 shows a comparison of the final values obtained theoretically against the experimental values. It can be observed that except the sizes of the active load (M_3 and M_4), the remaining sizes do hold good in the actual experimental values when compared to the final values.

Parameters	**Theoretical Values (in µm)**	**Simulated Values (in µm)**
W_1, L_1 (W_2, L_2)	**9, 0.2**	**16,0.6**
W_3, L_3 (W_4, L_4)	**.7, 0.3**	**16, 2.3**
W_5, L_5	**12, 0.3**	**25, 0.5**
W_6, L_6	**6, 0.2**	**67, 1.6**
W_7, L_7	**1.1,0.3**	**68, 0.5**

g.) Measuring Input Common-Mode Range (ICMR):

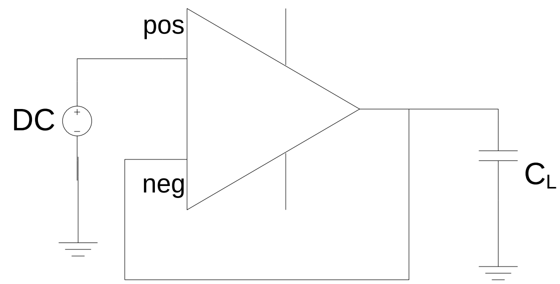


Figure 15 Unity-Buffer Op-amp

To measure the ICMR, the op-amp is connected as shown in Fig. 2. A DC simulation is run with the input DC Voltage swept from 0 to 1.8 V. At the output, Vout/Vin is plotted against Vin. The simulation for ICMR is in appendix A. There was no option of the ratio of Vout /Vin so the result is not clear in the figure.

- **Closed Loop OCMR:**

For closed-loop OCMR, the circuit is configured as shown in the figure 7.

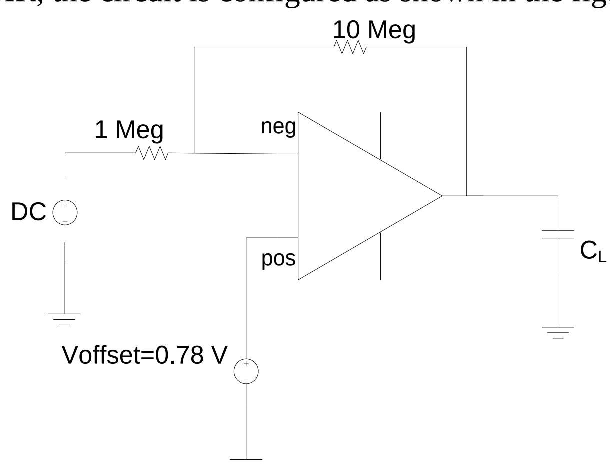


Figure 16 Closed Loop OCMR

Fig. 8 shows the OCMR outputs for open-loop (fig. 8a) and closed-loop configurations (fig. 8b). From Fig. 7b, it can be observed that the output range is from **1.6 to 0.160 V.**
To measure open-loop differential gain, bandwidth and phase margin, the

h.) Measuring Open-loop Differential Gain, Bandwidth and Phase Margin:

The open-loop differential gain, bandwidth and phase margin are measured from the "AC Gain and phase" plot with the op-amp connected as shown in the Fig. 9. ***The measured***

gain values are 65.34 dB with Unity Gain Bandwidth of more than 27.54 MHz and Phase margin of 65.78 degrees as shown in appendix A.

- ***Use of Compensation to improve Phase margin:***

Miller compensation technique was used to improve the phase margin and closed-loop stability of the op-amp. Frequency-comp

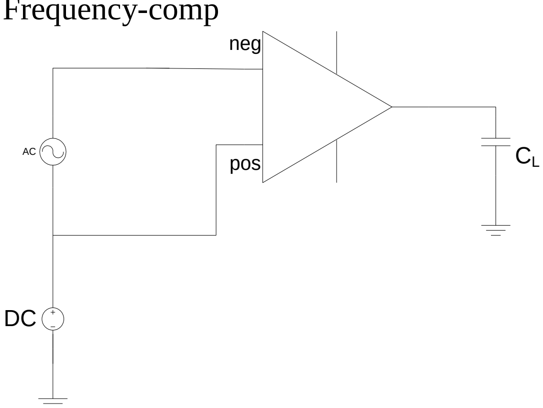


Figure 17 Measuring Open-loop Differential Gain

Figure 18 Open loop Differential Gain, BW and Phase Margin

i.) Measuring Power Dissipation:

For measuring the power dissipation, the op-amp is connected as a unity-gain buffer as shown in Fig. 2. It is found that the power dissipation is 1.8 V x 290.8 μA = 0.523 mW.

j.) Measuring Common-Mode Gain/CMRR:

For measuring common-mode gain, the circuit is connected as shown in Fig. 11. Both the inputs are connected together, and the input AC voltage and DC voltage are connected in series. This would yield the common-mode gain. To find CMRR, the common-mode gain is subtracted from the differential gain.

The CMRR is found to be 51 dB if the configuration in Fig. 1 is used. Hence, the circuit was modified to use a wide-range cascode current source as shown in appendix A. The transistor sizes are listed in Table 2. Using this configuration, the CMRR is found to be $Gdiff_{(dB)} - Gcm_{dB} = 83.55dB$ as shown in Appendix A.

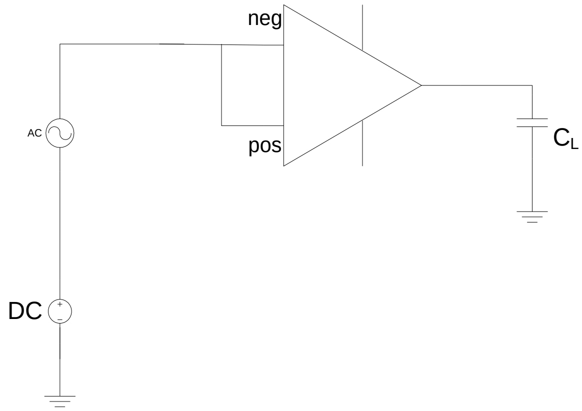
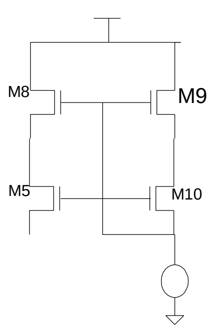


Figure 19 Measuring Common Mode Gain

- **Using wide –swing Cascode current mirror for CMRR compensation:**

By using the wide –swing Cascode current mirror the current gain error of the regular current mirror is less and the problem of voltage headroom does not exist. The mismatch of Vgs0 and Vgs5 creates some mismatches in the current gain and this will be reflected as a voltage offset at the output of amplifier. Using a cascade current mirror will improve the current gain but there will be voltage headroom problem, as there will be 4 transistors in series, wide-swing cascade current mirror will have the cascade output that has a higher r_{ds} at the output and therefore output offset voltage will be much less than previous case. This will make CMRR to go high in a very distinct way.

- Designing Wide-swing cascade current mirror.



for M9 to be in saturation M10 should be sized and biased in a way that :

vgs10 +vgs9-vth9 < Vb10 < vgs9+vth10
an in other word:
vgs10-vth10 <vth9

the transistors should be sized in a way that they satisfy this condition and also the node voltages of the current circuit remains close to the previous one.

Table IIIII Sizes for op-amp in wide-range cascode configuration

Circuit Elements	Sizes(w,l)um
M5	**25,0.5**
M8	**50,2**
M9	**160,2**
M10	**120,2**

k.) Measuring Slew Rate:

For measuring the slew rate, the circuit was connected as shown in Fig. 14. A transient simulation is run as shown and the frequency of the sinusoidal source is increased till the output waveform is unable to scale to the input voltage level of 1.1 V. The positive slew rate obtained is 17.99 V/μs and negative slew rate is 28.1 V/μs.

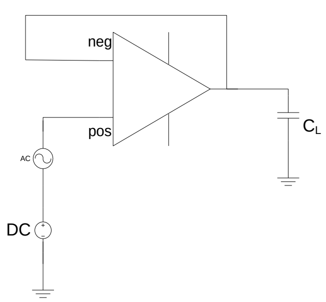


Figure 20 Measuring Slew Rate

4. Comparison between the two configurations

From comparing the two configuration of NMOS and PMOS, it can be observed that with PMOS configuration gaining higher Gain, Bandwidth, phase-margin and Slew-rate is less elaborative but the drawback of this configuration will be lower CMRR and Higher Power consumption and larger area.

But overall it can be concluded that PMOS is a better configuration for this application.

5. Conclusions

All the required specifications for the project were met once the cascode based current source was implemented. Here are the final results compared with the required specifications.

Parameters NMOS	***Simulated Values NMOS***	***Required specs NMOS***
Gain	65.56 dB	> 65 dB
Common-mode input range	0.557V to 1.656 V	0.7 V to 1.1 V
Common-mode output range (Closed loop and Open-loop)	0.16 to 1.573 V	0.25 V to 1.55 V
Common-Mode Rejection Ratio (low frequency)	94.58 dB ~ 95 dB	> 80 dB
Slew Rate	PSR: 10.81 V/µs NSR: 7.85 V/µs	> 10 V/µs
Bandwidth	20.893	> 20 MHz
Phase Margin	62.67 dB	> 60 º
Gain Margin	15.52 dB	> 10 dB
Parameters PMOS	***Simulated Values PMOS***	***Required specs PMOS***
Gain	65.34 dB	> 65 dB
Common-mode input range	0.557V to 1.656 V	0.7 V to 1.1 V
Common-mode output range (Closed loop and Open-loop)	0.16 to 1.6 V	0.25 V to 1.55 V
Common-Mode Rejection Ratio (low frequency)	83.55 dB	> 80 dB
Slew Rate	PSR: 17.99 V/µs NSR: 28.8 V/µs	> 10 V/µs
Bandwidth	27.4	> 20 MHz
Phase Margin	65.78 degrees	> 60 º
Gain Margin	20 dB	> 10 dB